MEDIUM · ADVANCED MATH

Advanced Math

08

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ADVANCED MATH

$$y = x^2 + 14x + 48$$

$$x + 8 = 11$$

The solution to the given system of equations is x, y. What is the value of y?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 GRID-IN ADVANCED MATH

The expression $90y^5 - 54y^4$ is equivalent to $ry^4 15y - 9$, where r is a constant. What is the value of r?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ADVANCED MATH

A rectangular volleyball court has an area of 162 square meters. If the length of the court is twice the width, what is the width of the court, in meters?

A. 9

B. 18

C. 27

D. 54

Q4 ADVANCED MATH

Blood volume, V_B, in a human can be determined using the equation $V_B = \frac{V_P}{1-H}$, where V_P is the plasma volume and H is the hematocrit (the fraction of blood volume that is red blood cells). Which of the following correctly expresses the hematocrit in terms of the blood volume and the plasma volume?

A. $H = 1 - \frac{V_P}{V_B}$

B. $H = \frac{V_B}{V_P}$

C. $H = 1 + \frac{V_B}{V_P}$

D. $H = V_B - V_P$

Q5 ADVANCED MATH

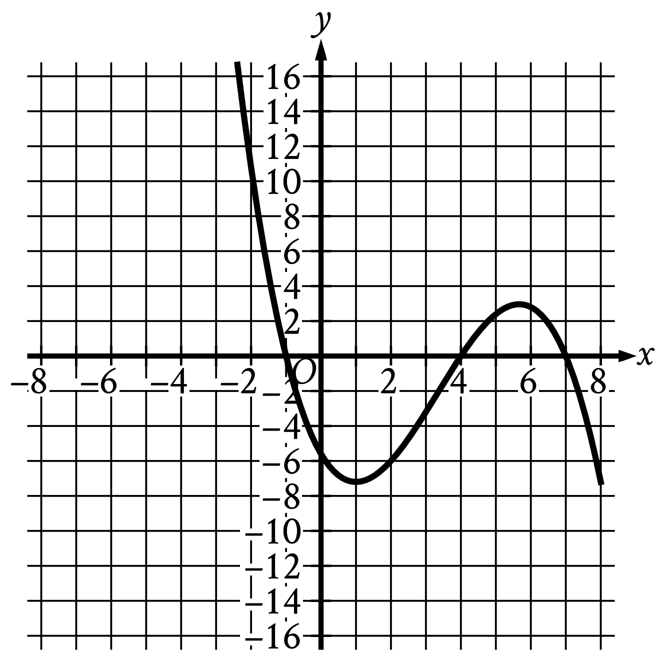
Which expression is equivalent to $d - 68d^2 - 3$?

A. $8d^3 - 14d^2 - 3d + 18$

B. $8d^3 - 17d^2 + 48$

C. $8d^3 - 48d^2 - 3d + 18$

D. $8d^3 - 51d^2 + 48$



- In quadrant 2 the curve falls sharply to cross the x axis at point (negative 1 comma 0).
- In quadrant 3 the curve falls sharply to cross the y axis at point (0 comma negative 5.5).
- In quadrant 4:
  - The curve falls gradually to a relative minimum at point (1 comma negative 7).
  - The curve rises sharply to cross the x axis at point (4 comma 0).
- In quadrant 1:
  - The curve rises sharply to a relative maximum at point (5.5 comma 3).
  - The curve falls sharply to cross the x axis at point (7 comma 0).
- In quadrant 4 the curve falls sharply.

The graph of $y = fx$ is shown, where the function f is defined by $fx = ax^3 + bx^2 + cx + d$ and a, b, c, and d are constants. For how many values of x

does $fx = 0$?

A. One

B. Two

C. Three

D. Four

Q7

ADVANCED MATH

$$x^2 = 6x + y$$
$$y = -6x + 36$$

A solution to the given system of equations is (x,y). Which of the following is a possible value of xy ?

A. 0

B. 6

C. 12

D. 36

Q8 ADVANCED MATH

Which expression is equivalent to $a^{\frac{11}{12}}$, where $a > 0$?

A. $\sqrt[12]{a^{132}}$

B. $\sqrt[144]{a^{132}}$

C. $\sqrt[121]{a^{132}}$

D. $\sqrt[11]{a^{132}}$

x	f(x)
0	5
1	$\frac{5}{2}$
2	$\frac{5}{4}$
3	$\frac{5}{8}$

The table above gives the values of the function f for some values of x. Which of the following equations could define f ?

A. $f(x) = 5(2^{x+1})$

B. $f(x) = 5(2^{x})$

C. $f(x) = 5(2^{-(x+1)})$

D. $f(x) = 5(2^{-x})$

Q10 ADVANCED MATH

An oceanographer uses the equation $s = \frac{3}{2}p$ to model the speed s, in knots, of an ocean wave, where p represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave?

A. $p = \frac{2}{3}s$

B. $p = \frac{3}{2}s$

C. $p = \frac{2}{3} + s$

D. $p = \frac{3}{2} + s$